



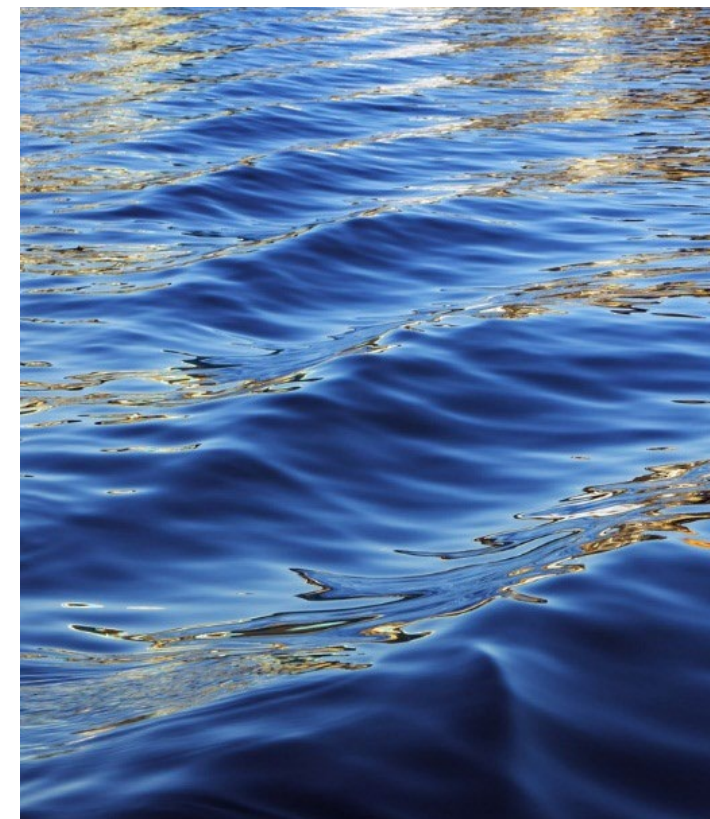
Risk Management

Random Risks

Disruption



European University Alliance for Global Health



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Universidad
de Alcalá



Learning objectives

- Qualify disruptive events.
- Model Ripple Effect.
- Set management strategies for disruption.
- Use cases and examples



Disruptive events in the Supply Chain

A disruptive event in the supply chain is any unexpected occurrence that significantly interrupts or alters the normal flow of goods, services, information, or finances between different links of the supply chain, leading to negative impacts on costs, timelines, quality, or availability.

- **Internal** (e.g., production failures, logistical errors)
- **External** (e.g., natural disasters, geopolitical conflicts, pandemics, strikes, cyberattacks).



Features of Disruptive events

Severity

Scope
(broadness)

Duration

Origin

Frequency
(probability)

Early
Detection

Propagation
Speed

Economic
Impact

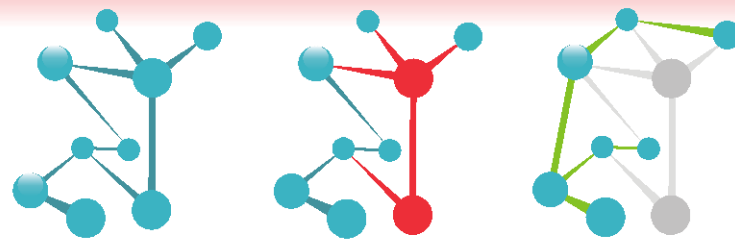
Recoverability
(resilience)



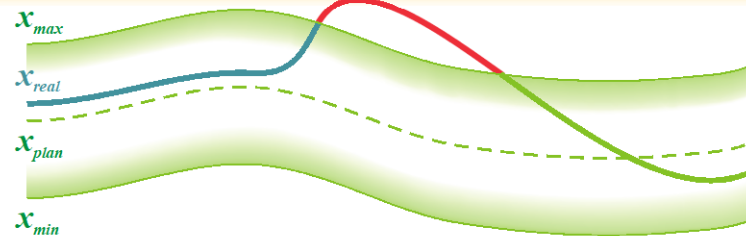
Ripple effect in the Supply Chain

Describes the impact of a disruption in SC performance, the propagation of disruptions, and the disruption-based scope of changes to SC structures and parameters.

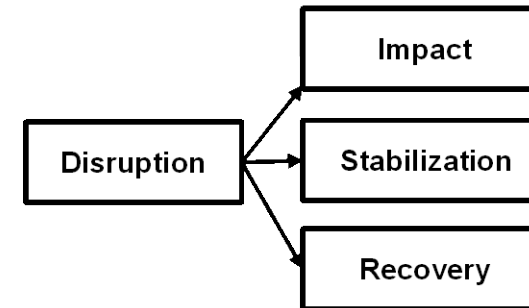
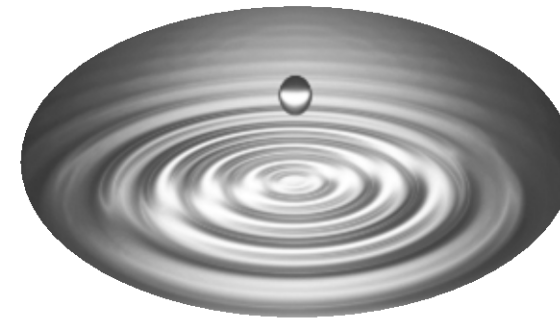
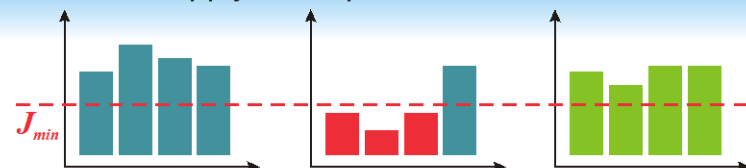
supply chain structure dynamics control and adaptation



operations execution and stability control



supply chain performance control



Ivanov D., Dolgui, A., Sokolov, B. (2014) The Ripple-Effect in Supply Chains: Trade-off “efficiency-flexibility-resilience” in supply chain disruption management. International Journal of Production Research, 52:7, 2154-2172.



Some Examples

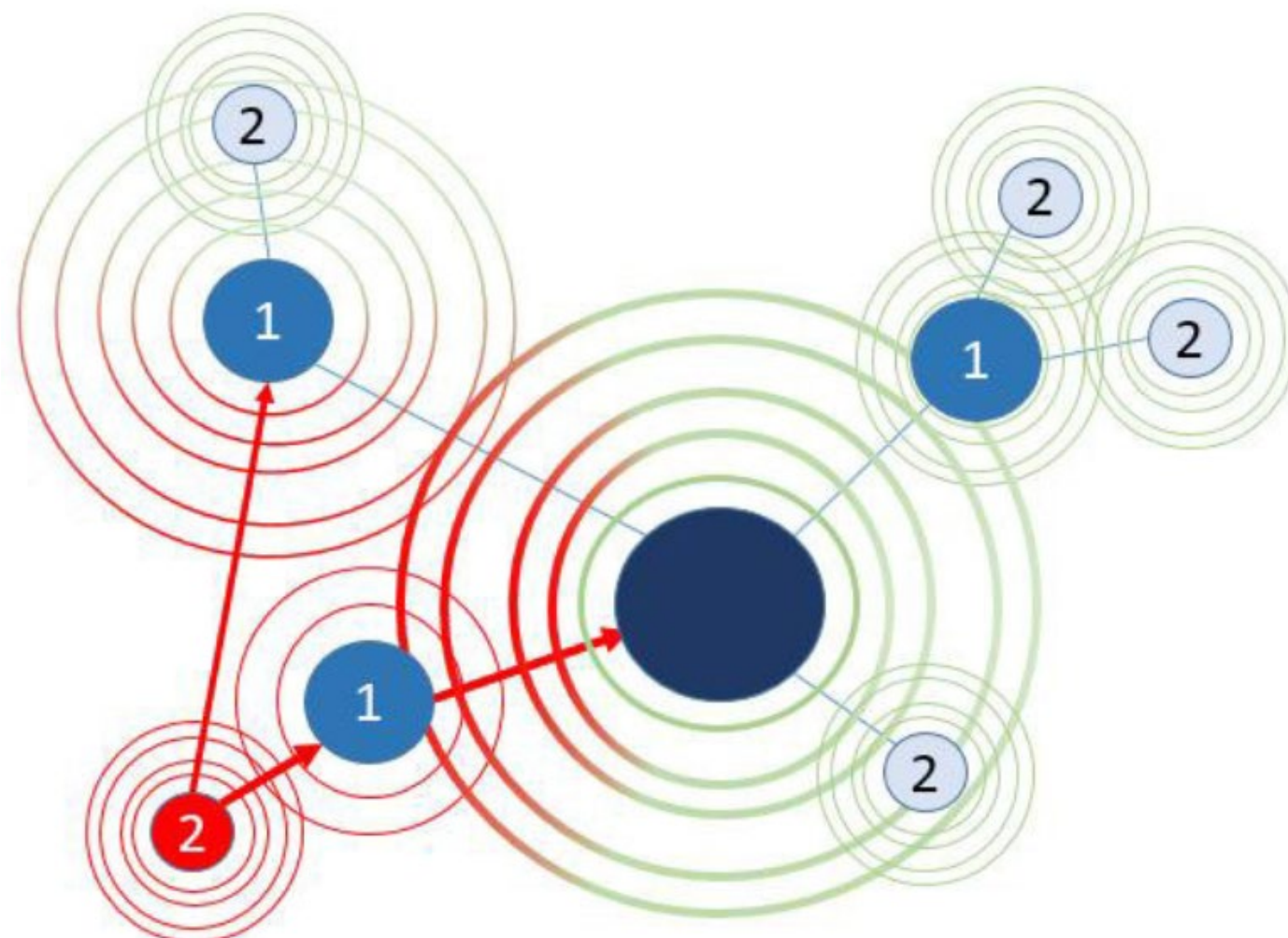
Factor	Example	Impacts
Terrorism Piracy	September 11 Somali, 2008	Five Ford plants have been closed for a long time Breaks in many supply chains
Natural disasters	Earthquake in Thailand, 1999 Flood in Saxony, 2002 Earthquake in Japan, 2007	Apple computers' production in Asia has been paralysed Significant production decrease at VW, Dresden Production breakdown in Toyota's supply chains amounted to 55.000 cars
	Hurricane Katrina, 2006	This storm halted 10%-15% of total US gasoline production, raising both domestic and overseas oil prices
	Earthquake and tsunami in Japan, 2011	Massive collapses in global automotive and electronics supply chains; Toyota lost its market leadership position
	Floods in Chennai, India in 2015	Production of academic literature has been stopped at many international publishing houses



More examples

Man-made disasters	Explosion at BASF plant in Ludwigshafen in 2016	15% of raw materials were missing for the entire supply chain Production of some products at BASF has been stopped for many weeks
	Fire at distribution center of e-commerce retail company ASOS in 2005	Delivery stop for a month
	A fire in the Phillips Semiconductor plant in Albuquerque, New Mexico in 2000	Phillips's major customer, Ericsson, lost \$400 million in potential revenue
Political crises	"Gas" crisis 2009	Breaks in gas supply from Russia to Europe, billions of losses to GAZPROM and customers
Strikes	Strikes at Hyundai plants in 2016	Production of 130,000 cars has been affected
Legal contract disputes	Volkswagen and Prevent Group contract dispute in summer 2016	Six German factories face production halt on parts shortage; 27,700 workers are affected, with some sent home and others moved to short-time working

Ripple Effect: impact propagation in the SC



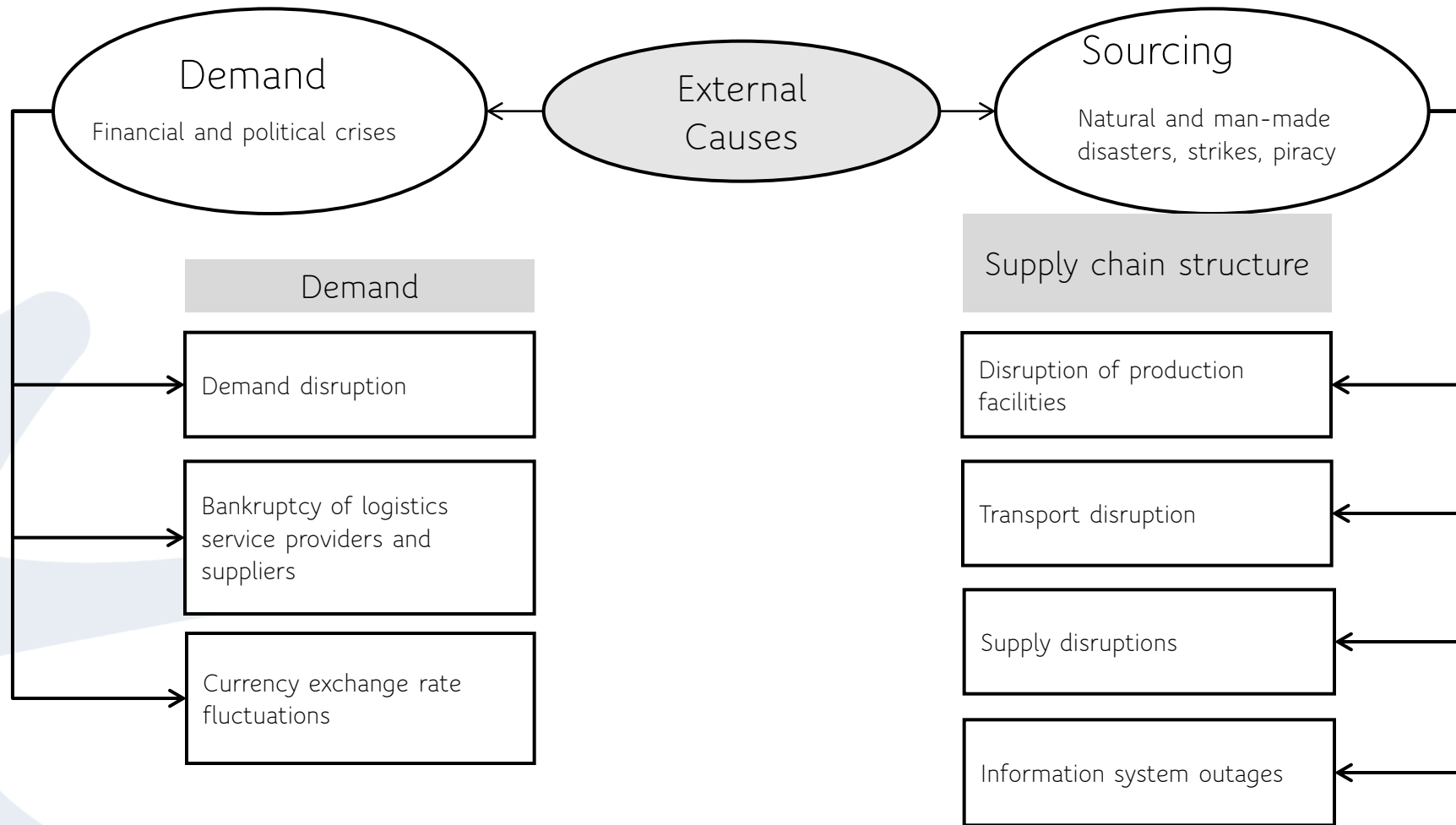


Ripple vs. Bullwhip

	RIPPLE Effect	BULLWHIP Effect
Risks	Disruptions (e.g. Fire)	Operational (e.g. demand fluctuation)
Affected areas	Structure and critical parameters (unavailability, lost sales, etc).	Operations: lead-time, inventory, profitability, etc.
Recovery	Medium-long term. Considerable effort and investment. Strategy changes.	Short-term coordination to balance demand and delivery.
Impact on efficiency	Impact on annual accounts and profits.	On-time and on-going efficiency, in the form of stock-out, cost overruns, etc.

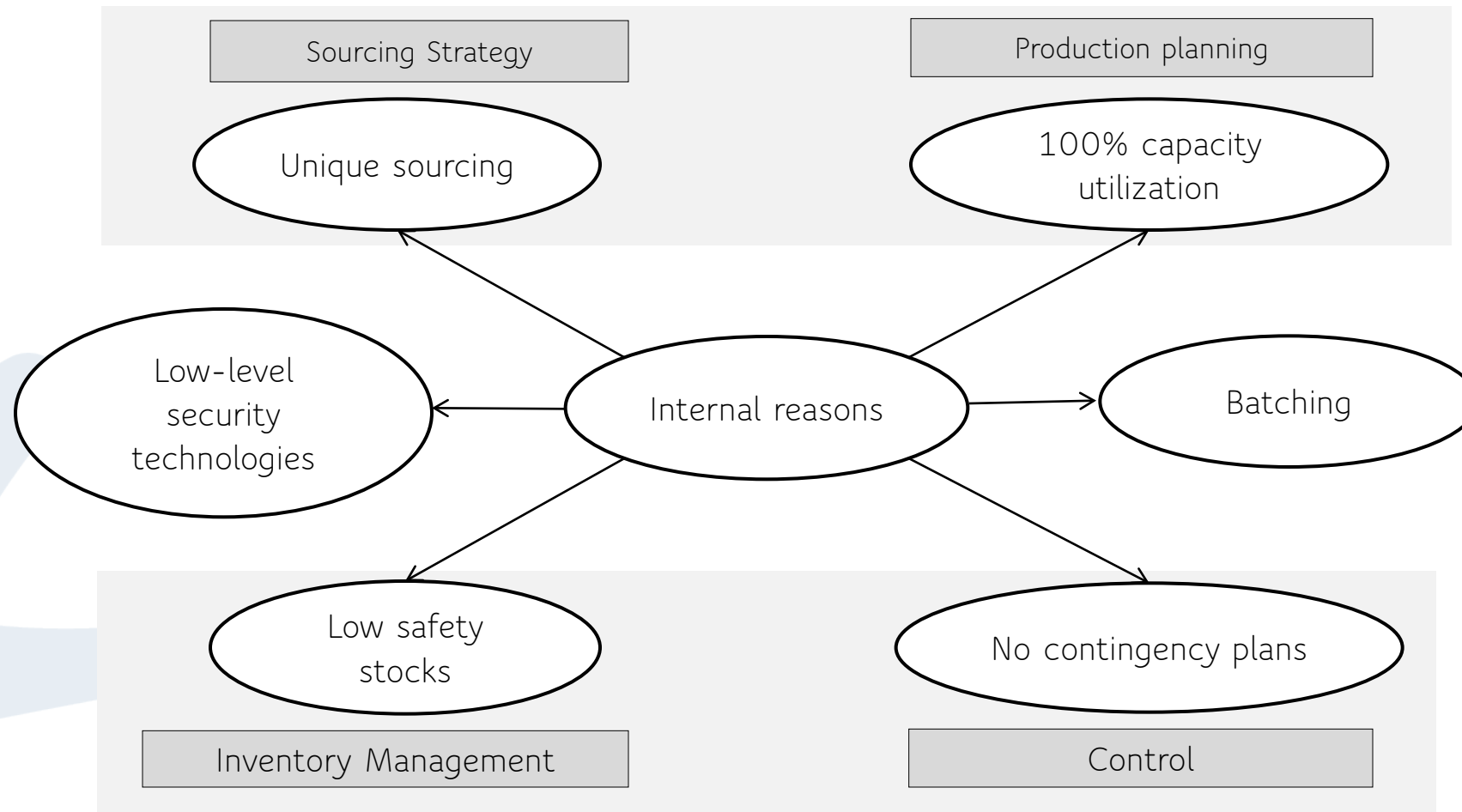


External causes



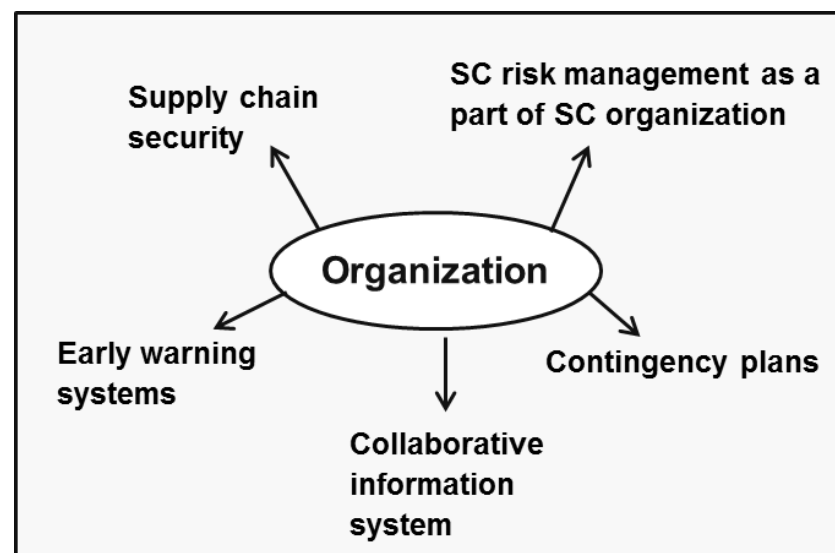
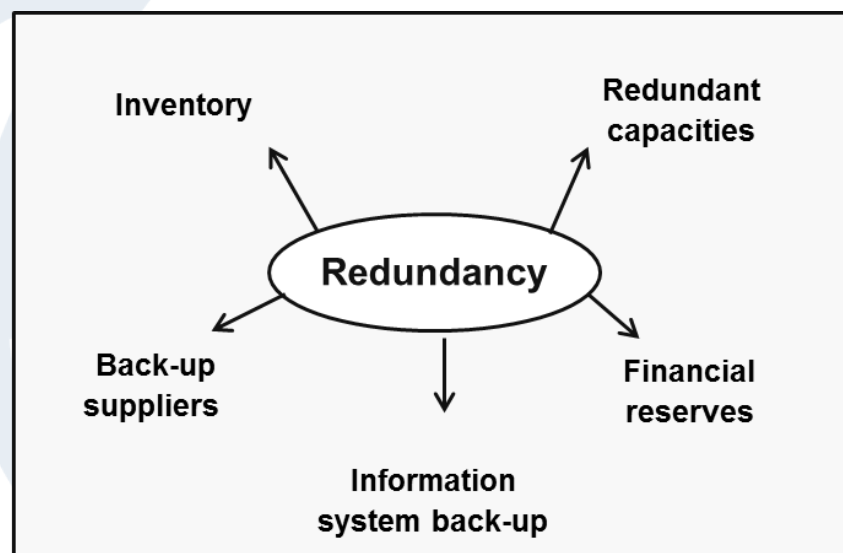
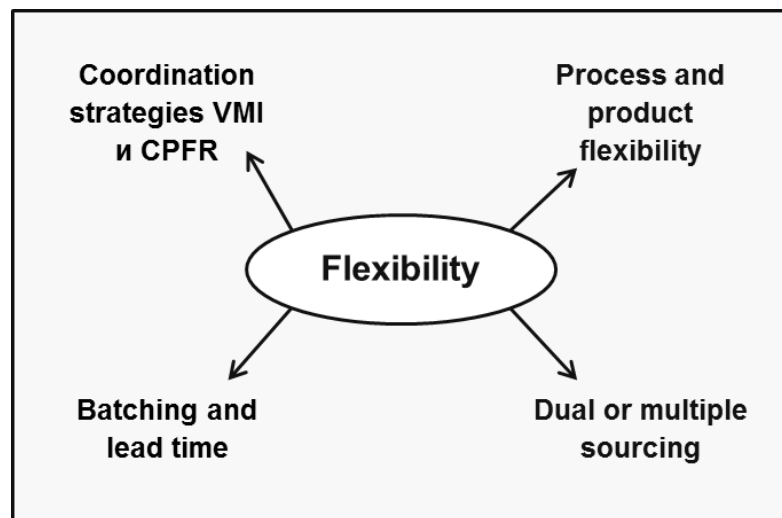


Internal Causes



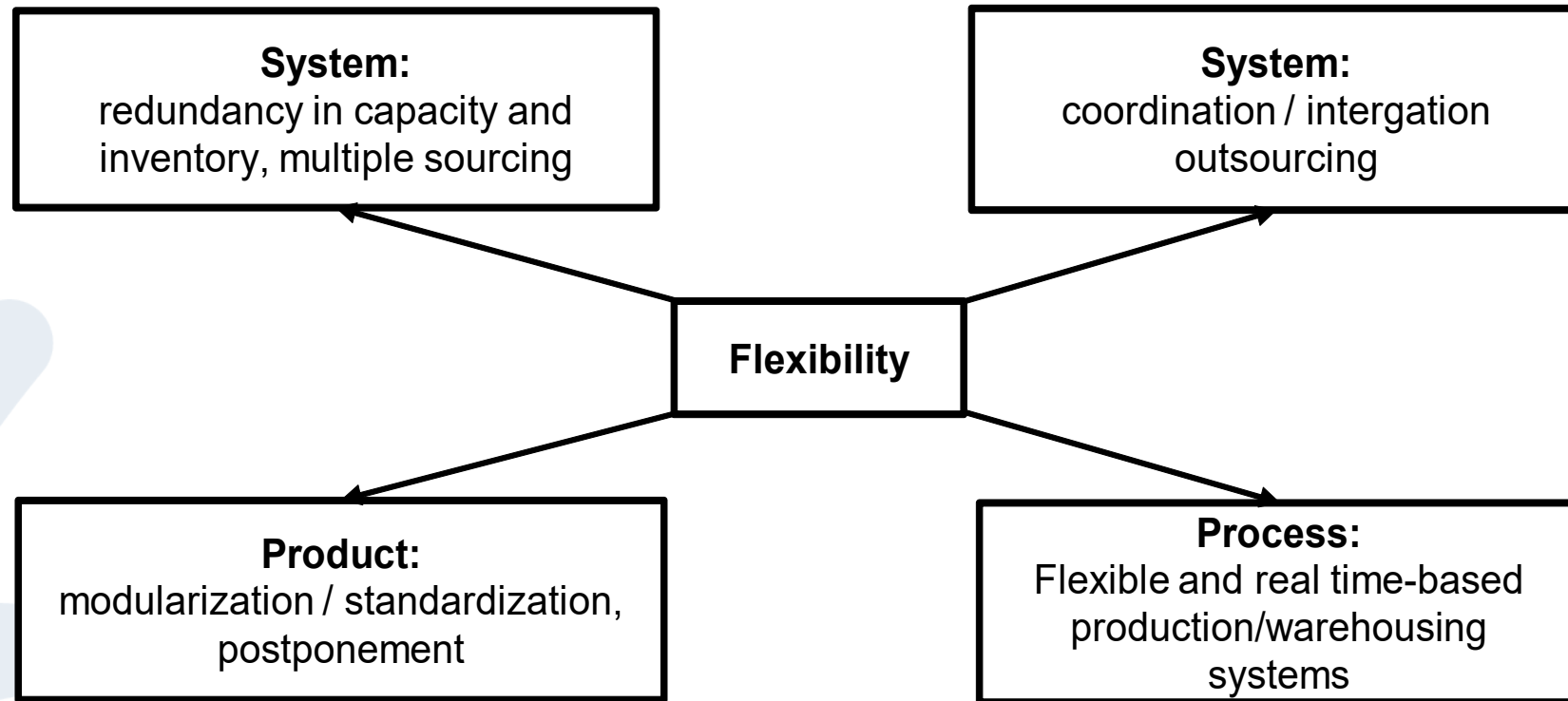


Preventive Countermeasures



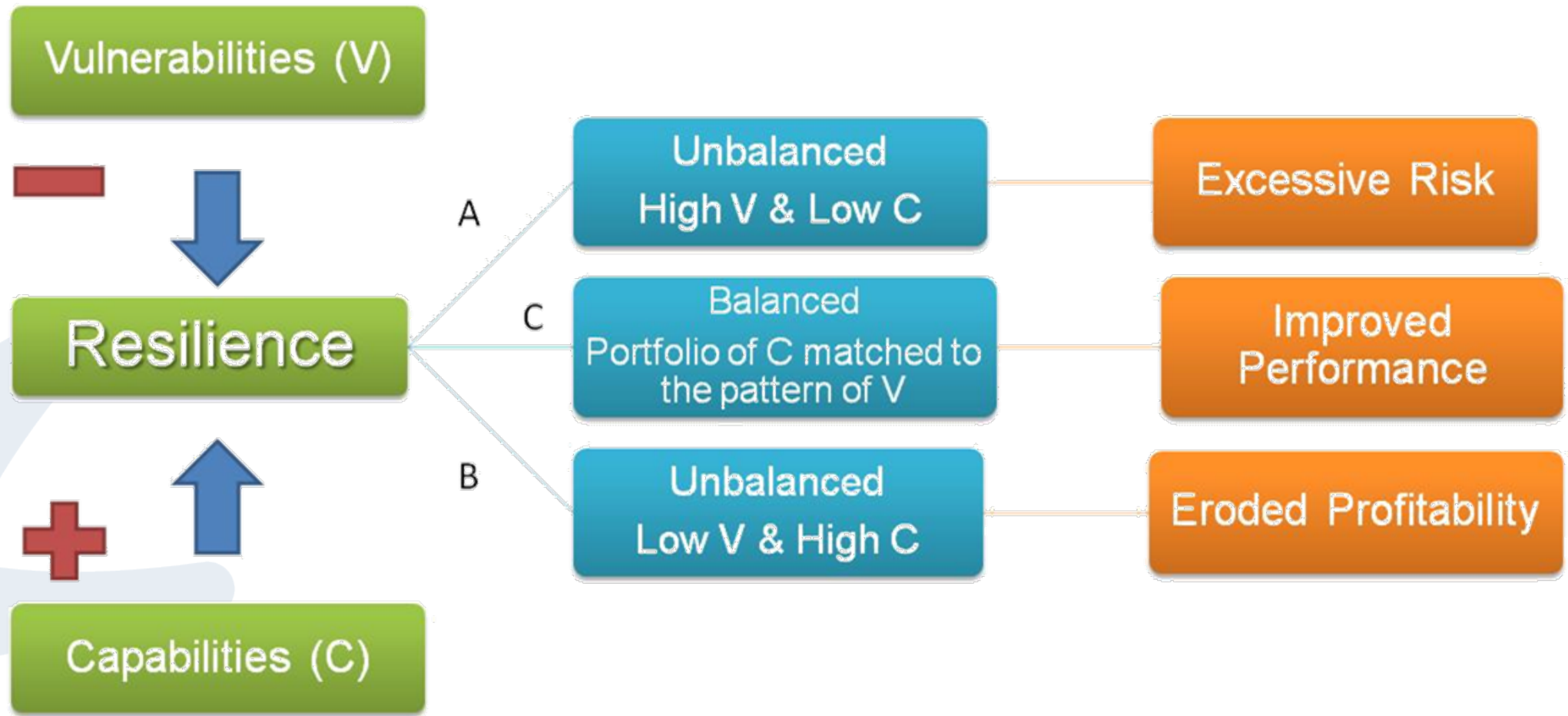


Flexibility is a critical feature





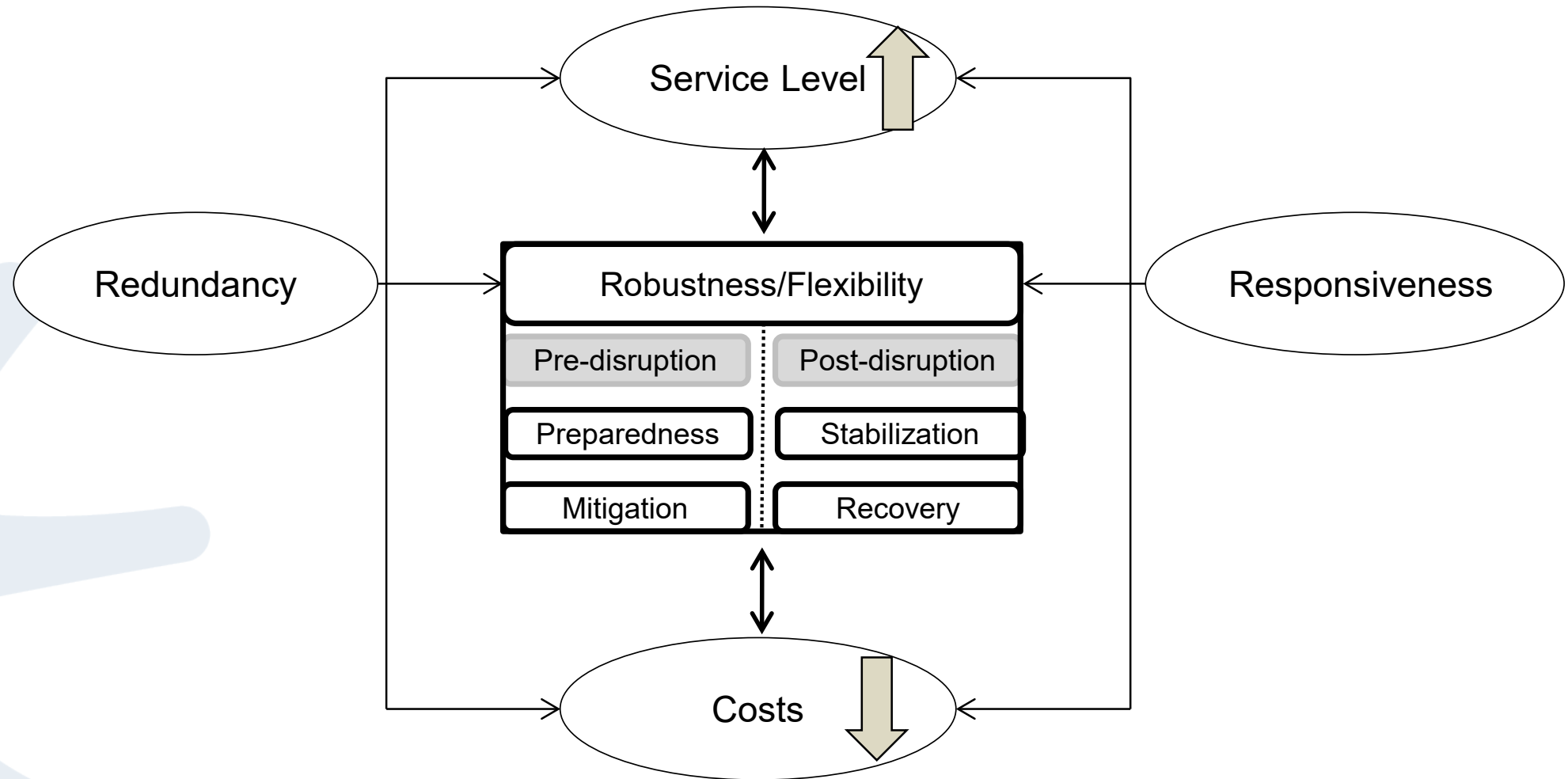
Resilience depends on the VOLUME $\leftrightarrow$ COST balance



Source: Adopted by Pettit et. al. 2010

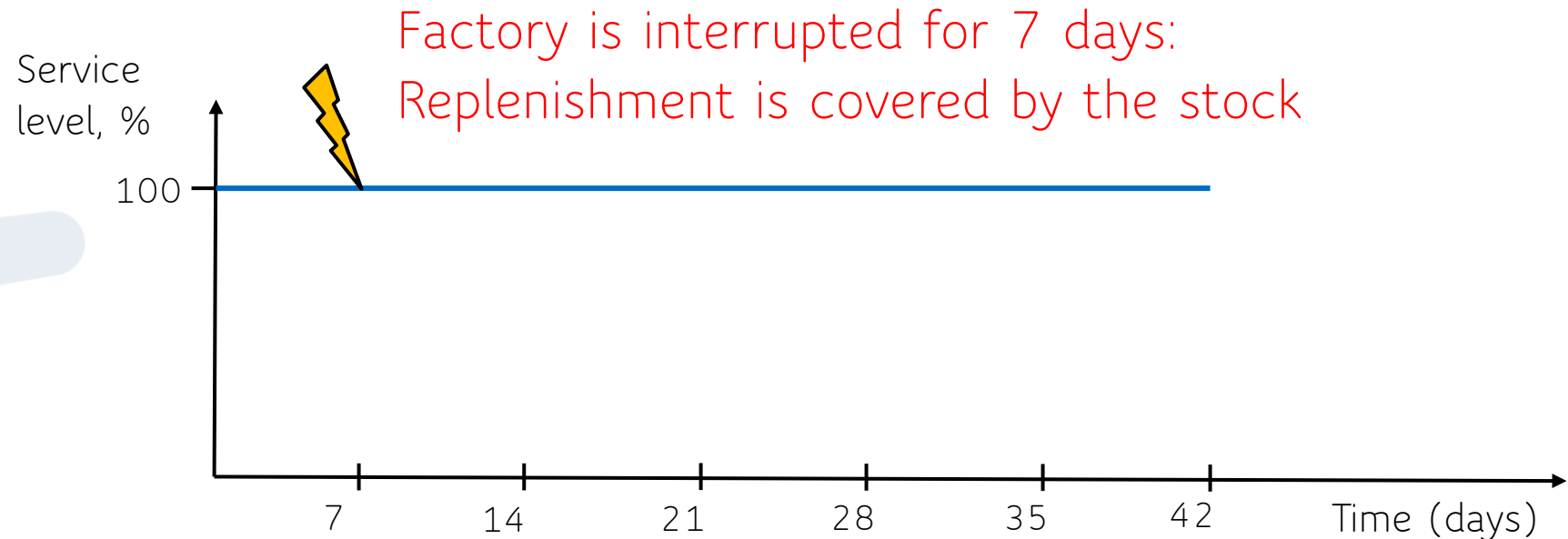
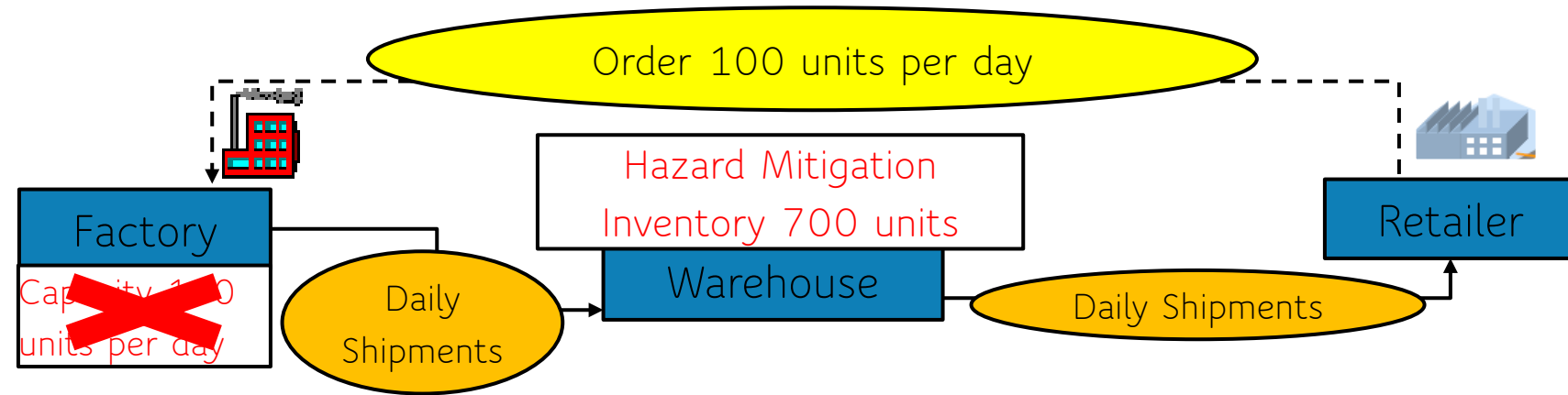


Resilience framework



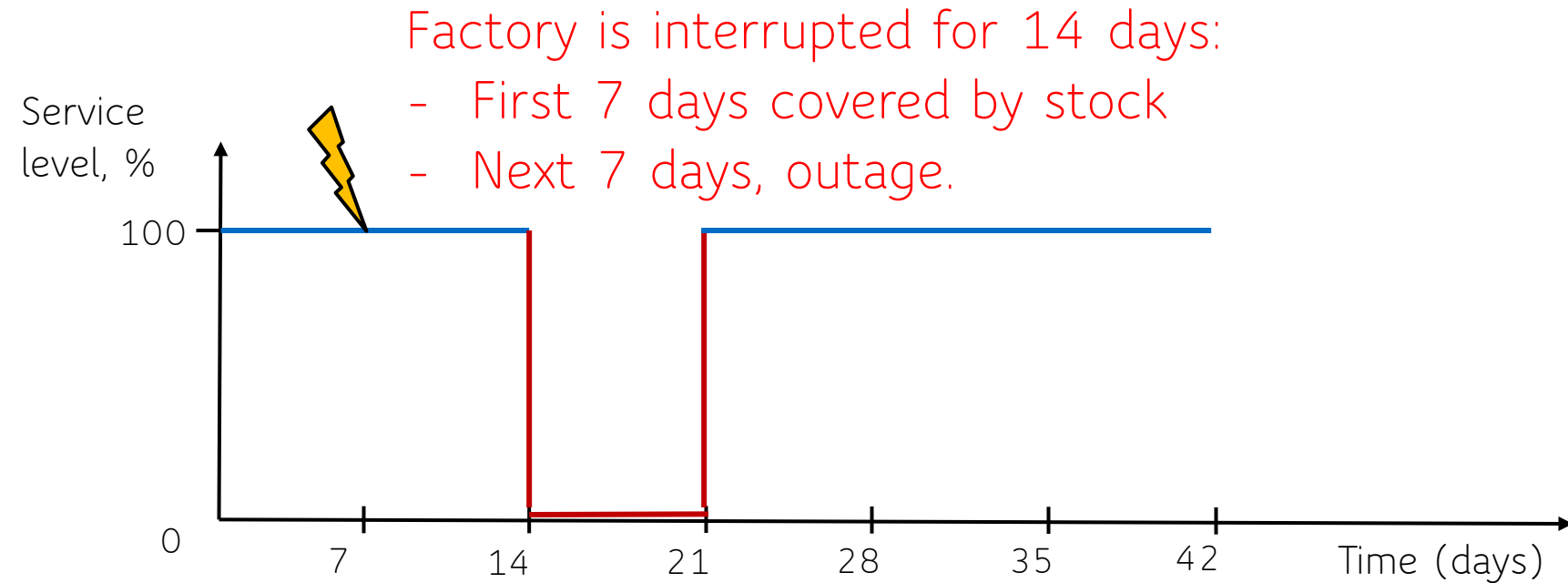


A proactive protection strategy





The proactive strategy has limitations...



- It could be covered also with a back-up factory or DC.

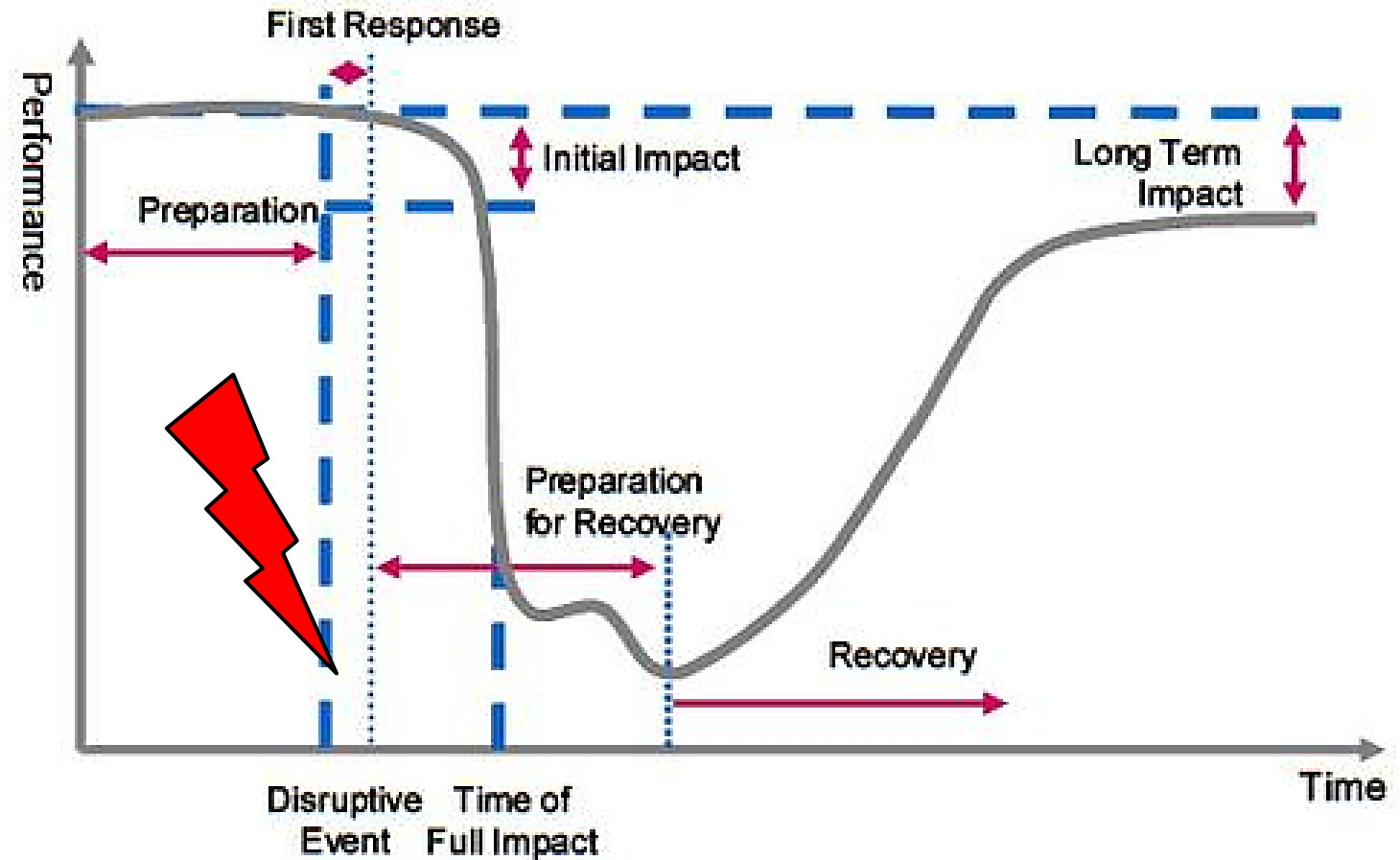
Is it effective to increase the stock to cover wider impacts?

The increased safety stock and backup factory increase costs!



Recovery time and impacts

- Survival time vs recovery time
- Recovery costs vs recovery time



Adapted from: Sheffi Y., Rice J.B. (2005). A Supply Chain View of the Resilient Enterprise. MIT Sloan Management Review

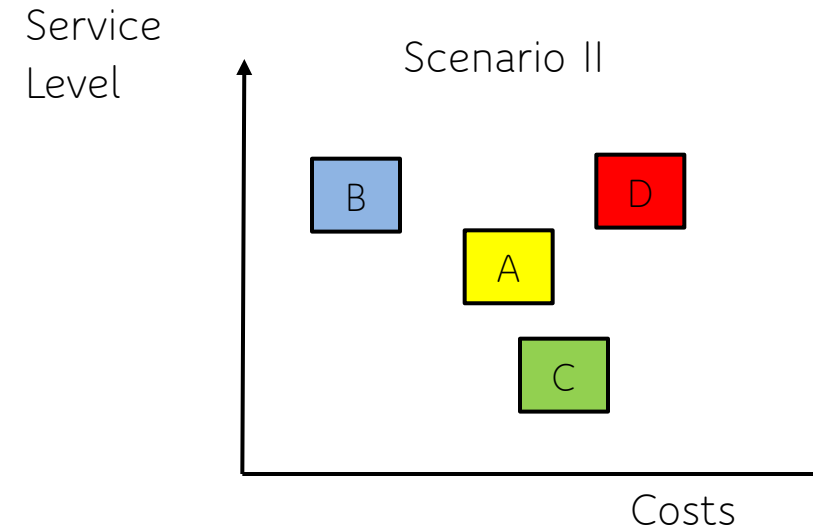
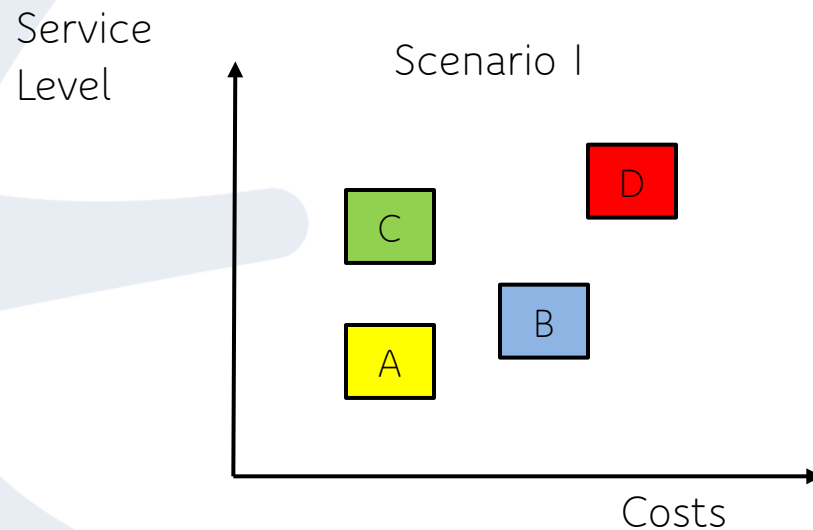


Multiple events need to be covered

We are interested in investigating the impact on supply chain performance of different combinations of disruptions, proactive mitigation strategies, and reactive recovery policies.

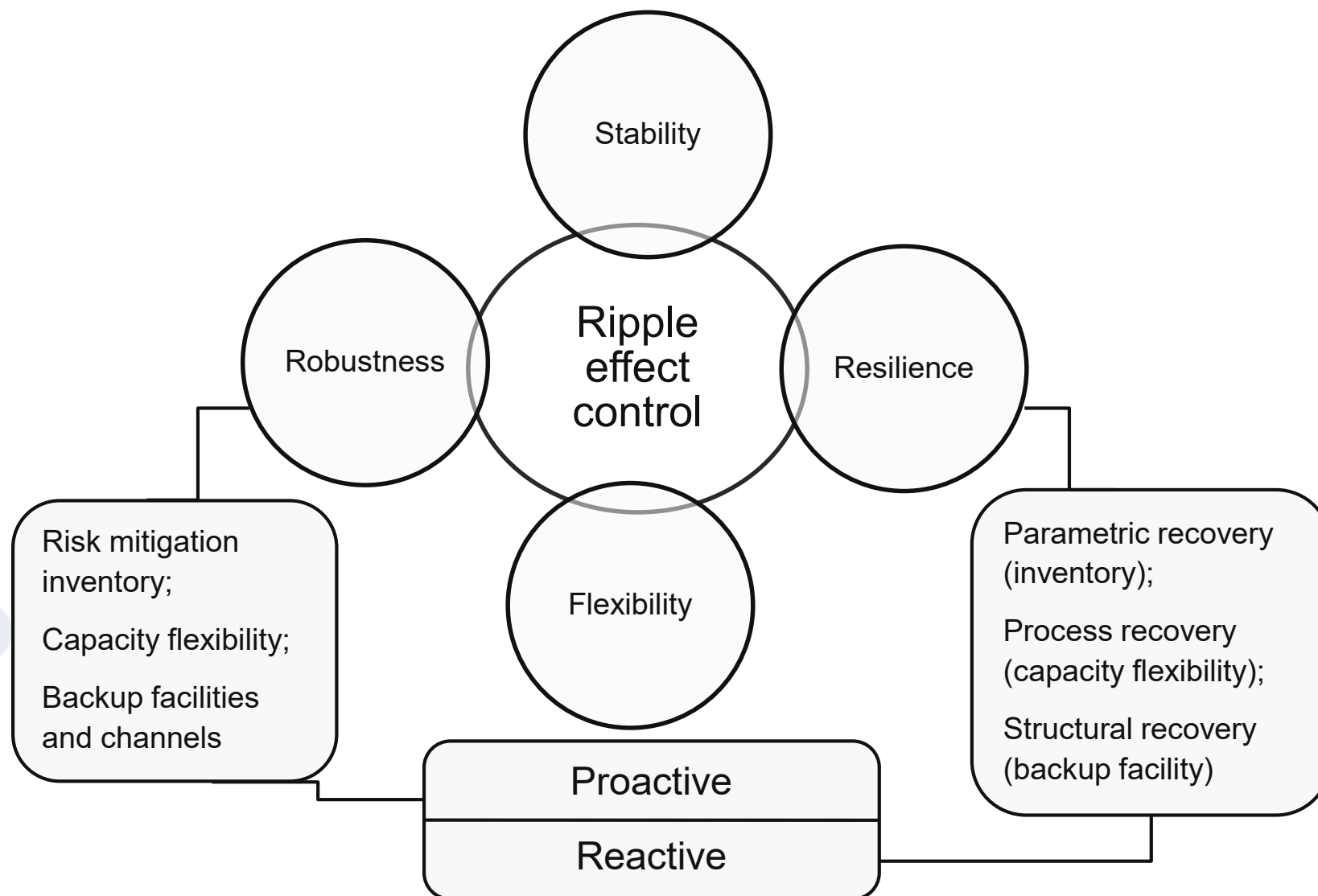
Example:

- Two Disruption Scenarios I and II
- Four Proactive and Reactive Policy Combinations A, B, C, and D





Integrated supply chain for resilient service





Session Recap

- Qualify disruptive events.
- Model Ripple Effect.
- Set management strategies for disruption.
- Use cases and examples